

CPR 1 MODULE OBJECTIVES				
Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 1	ANAT	Mediastinum	SOM.MK.II.BPM1.3.CPR.2.ANAT.1363	Describe the subdivisions of the mediastinum and their boundaries.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1364	List the structures found in the superior, anterior, middle and posterior mediastinum and their relationships to one another.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1365	Describe the relationships of the esophagus, trachea and thoracic duct in the superior mediastinum.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1366	Describe the typical branching pattern of the aortic arch in the superior mediastinum.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1367	Describe the subclavian artery and its branches.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1368	Explain what clinical impairments could be caused by the presence of a cervical rib.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1369	Describe the relationships of the phrenic nerves to the aortic arch and the root of the lung.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1370	Describe the relationships of the vagus nerves to the aortic arch, the root of the lung and the esophagus.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1371	Describe the course of the right and left recurrent laryngeal nerves in the thorax.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1372	List the structures that are found at the plane of Ludwig.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1373	Describe the four normal constrictions of the esophagus.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1374	Describe the arterial supply, innervation, venous and lymphatic drainage of the thoracic and abdominal portions of the esophagus.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1375	Describe the azygos system of veins and their typical pattern of drainage.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1376	Describe the origin of the greater, lesser and least splanchnic nerves.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1377	List the normal openings in the diaphragm and their approximate vertebral level.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1378	Describe the anatomical location, mechanism and clinical sequences of thoracic duct laceration.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1379	Describe the clinical signs, causes and presentation of thoracic outlet syndrome
Lecture 2	ANAT	Clinical Anatomy of the Heart	SOM.MK.II.BPM1.3.CPR.2.ANAT.1380	Describe the location of the heart in the mediastinum.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1381	Describe the path of blood flow through the heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1382	Differentiate between the systemic and the pulmonary circulation.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1383	List the pericardial coverings of the heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1384	Describe the extent and attachments of the fibrous pericardium (pericardial sac).
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1385	Describe the innervation and arterial supply of the fibrous pericardium.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1118	Describe the clinical significance of the transverse pericardial sinus
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1387	List the surfaces and margins of the heart and the chambers/structures that contribute to them.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1388	Describe the anatomy of the great vessels, heart chambers and their relationships to one another and to adjacent structures.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1389	Describe the importance of the aortic sinuses (of Valsalva) for continuous myocardial perfusion.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1390	Describe the surface projections of the heart and the heart valves.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1391	Identify the auscultation sites for heart valve sounds and explain why these auscultation points differ from the anatomical location of the valves.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1392	Describe the arterial supply and venous drainage of the heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1393	Explain which part of the myocardium will be affected if any of the coronary arteries is occluded.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1394	Describe the concept of left and right dominance as it relates to the blood supply of the heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1395	Describe the sympathetic, parasympathetic and sensory innervation of the heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1396	Describe the conducting pathway of the heart
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1397	Describe the course of the pain fibers from the heart to the CNS
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1398	Differentiate between the pathways used by the two types of visceral afferent fibers (carrying pain versus physiologic sensation) from the heart to the CNS.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1399	Explain why the pain of a myocardial infarction maybe referred to the upper limb.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1400	Describe the intercostobrachial nerve and its clinical significance in relation to the upper limb and myocardial infarction.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1401	Explain how cardiac pain differs from intercostal pain/pleural pain /bronchial pain
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0079	List the major functions of the cardiovascular system.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0080	Explain the organization of the circulatory system.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0081	Explain why the output from the left heart must be more or less the same as the output from the right heart.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0082	Define systole and diastole.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0083	State the key features that distinguish the pulmonary and systemic circulations
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0084	Describe the role of the hepatic portal circulation in providing flow between capillaries and by-passing the heart.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0085	Define cardiac output (CO) and stroke volume (SV). Explain how CO and SV are calculated and be able to list normal value ranges.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0086	Write down the basic law of flow.

DLA 1	PHYS; HCB	Introduction to the Cardiovascular System	SOM.MK.II.BPM1.3.CPR.2.PHYS.0087	Apply the law of flow to explain the pressure changes seen across the circulation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0088	Explain why velocity of flow is slowest in capillaries.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0089	Describe the three main classes of blood vessels and explain how their morphology is suited to their functions as capacitance, resistance and exchange vessels.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1101	List the three layers (tunics) forming the walls of arteries and veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1102	Identify and describe the structure of the tunica intima.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1103	Identify and describe the structure of tunica media.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1104	Identify and describe the structure of tunica adventitia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1105	State the function(s) of vaso vasorum.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1106	State the function(s) of nervi vascularis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1107	State the origin of the elastic of the tunica media.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1108	State the origin of the reticular fibers of the tunica media.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1109	Give the function of the heart valves.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1110	Describe the histological structure of the heart valves.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1111	State the origin of the extracellular matrix (fibers and ground substance) found in the blood vessel wall.
Lecture 3	HCB	Histology of the Cardiovascular System	SOM.MK.I.BPM1.3.CPR.1.HCB.1112	List the cells and fibers found in the various tunics.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1113	List the 3 layers of the heart.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1114	Describe the structure and function of the 3 layers of the heart (epicardium, myocardium, endocardium).
			SOM.MK.I.BPM1.3.CPR.1.HCB.1115	Give the function(s) of Purkinje fibers.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1116	Sate the localization of Purkinje fibers.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1117	Describe the microscopic structure of Purkinje fibers.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1118	Discuss the functions of vascular endothelial cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1119	Describe the microscopic structure of vascular endothelial cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1120	List 3 substances released by endothelial cells that prevent thrombus formation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1121	List at least 5 vasoactive substances released by the endothelial cells
			SOM.MK.I.BPM1.3.CPR.1.HCB.1122	Explain how these vasoactive substances influence blood flow.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1123	Describe the process of "endothelial activation".
			SOM.MK.I.BPM1.3.CPR.1.HCB.1124	Discuss the clinical significance of "endothelial activation".
			SOM.MK.I.BPM1.3.CPR.1.HCB.1125	State the functional significance of Weibel Palade bodies
			SOM.MK.I.BPM1.3.CPR.1.HCB.1126	Briefly describe the structure of Weibel Palade bodies.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1127	Explain the significance of growth factors (VEGF) produced by endothelial cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1128	Give the function of large (elastic) arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1129	Describe the microscopic structure of large (elastic) arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1130	Give the function of medium (muscular) arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1131	Describe the microscopic structure of medium (muscular) arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1132	Give the function of small arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1133	Describe the microscopic structure of small arteries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1134	Give the function(s) of arterioles.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1135	Describe the microscopic structure of arterioles.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1136	Describe the microsocpic structure of the three types of capillaries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1137	Identify the three types of capillaries in electron photomicrographs.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1138	Give the function of venules (postcapillary and muscular).
			SOM.MK.I.BPM1.3.CPR.1.HCB.1139	Describe the microscopic structure of venules (postcapillary and muscular).
			SOM.MK.I.BPM1.3.CPR.1.HCB.1140	Give the function of small veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1141	Describe the microscopic structure of small veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1142	Give the function of medium veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1143	Describe the microscopic structure of medium veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1144	Give the function of large veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1145	Describe the microscopic structure of large veins.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1146	Describe the microscopic structure of lymphatic capillaries.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1147	Describe the microscopic structure of lymphatic vessels.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1148	Identify the layers of the vessel wall involved in "arteriosclerosis" and "atherosclerosis".
			SOM.MK.I.BPM1.3.CPR.1.HCB.1149	Define the term "aneurysm".
			SOM.MK.I.BPM1.3.CPR.1.HCB.1150	Identify the layer of the vessel wall affected that lead to the formation of aneurysm.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1151	Identify the layers of the vessel wall affected in aortic dissection.
Lecture 4	PHYS	Cardiac Cycle	SOM.MK.II.BPM1.3.CPR.2.PHYS.0090	Illustrate and explain, in a correct temporal relationship, the phases of the cardiac cycle (atrial systole, isovolumetric contraction, rapid ventricular ejection, reduced ventricular ejection, isovolumetric relaxation, rapid ventricular filling, reduced ventricular filling/diastasis) and the relationship of each phase with the left ventricular pressure curve, the left ventricular volume curve, the left atrial pressure curve, the aortic pressure curve, and the heart sounds (S ₁ , S ₂ , S ₃ , S ₄).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0091	Explain the ventricular volume-pressure relationship during the different phases of the cardiac cycle in the pressure-volume loop, and the points in the graph that correspond to the the mitral valve closure (S ₁), the mitral valve opening, aortic valve closure (S ₂), the aortic valve opening, S ₃ sound, S ₄ sound, end diastolic volume, and end systolic volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0092	Compare the timing of the different phases of the cardiac cycle in the left side versus the right side the heart
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0093	List the role of factors that contribute to the formation of the turbulent flow
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0094	Describe the timing and causes of the four heart sounds.

			SOM.MK.II.BPM1.3.CPR.2.PHYS.0095	Identify the characteristics of the heart murmurs associated with mitral stenosis, mitral regurgitation, mitral valve prolapse, aortic stenosis, aortic regurgitation, persistent ductus arteriosus, and ventricular septal defect
Lecture 5	ANAT	Imaging of the Heart and Mediastinum	SOM.MK.II.BPM1.3.CPR.2.ANAT.1351	Identify the major anatomical features of the mediastinum, heart and great vessels in radiographic images, CT and MRI.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1421	Describe the anatomical location, mechanism and clinical sequences of pericarditis, pericardial friction rub and pericardial effusion. (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1422	Define cardiac tamponade (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1423	Describe the anatomical explanation of the three cardinal signs of cardiac tamponade (Beck's triad) (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1424	Describe the paraxiphoid and apical approach to pericardiocentesis for the alleviation of cardiac tamponade. (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1425	Describe the pattern of the venous blood flow from the upper limb to the right atrium. (SG)
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1426	Explain why a swallowed sharp object might pierce the heart and into what chamber it could enter. (SG)
DLA 2	ANAT	Development of the Heart	SOM.MK.II.BPM1.3.CPR.2.ANAT.1402	Describe the formation of the oblique and the transverse pericardial sinus.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1403	List the subdivisions of the tubular heart.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1404	Describe the difference between dextrocardia and situs inversus
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1405	Describe the partitioning of the atrioventricular canal.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1406	Describe the formation of the right and left atria.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1407	Describe the development of the interatrial septum.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1408	Describe the fate of the sinus venosus
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1410	Describe the partitioning of the common ventricle.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1411	Describe the partitioning of the bulbus cordis and truncus arteriosus by the aorticopulmonary septum.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1413	Describe the development of the heart valves.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1414	List the cardiac structures derived from the neural crest cells.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1415	Explain the fate of vitelline, umbilical and cardinal vessels. (Reading assignment)
DLA 3	ANAT	Aortic Arches	SOM.MK.II.BPM1.3.CPR.2.ANAT.1346	Summarize the major events in the transformation of the pharyngeal arch system into the adult arterial pattern.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1347	Outline the embryologic basis of double aortic arch, right aortic arch, and anomalous right subclavian artery
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1348	Discuss the relatively common aortic arch anomalies, including patent ductus arteriosus and coarctation of the aorta.
Lecture 6	ANAT	Clinical Embryology of the Heart	SOM.MK.II.BPM1.3.CPR.2.ANAT.1416	Describe fetal circulation and discuss the changes that occur at birth.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1418	List left-to-right shunts in congenital heart defects and explain why they are generally acyanotic.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1409	Describe five types of atrial septal defects.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1412	Describe the two types of ventricular septal defect.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1417	List the most common heart defects in Down's syndrome.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1419	List right-to-left shunts in congenital heart defects and explain why they are generally cyanotic.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1420	Describe the developmental mechanisms underlying ectopia cordis, persistent truncus arteriosus, and transposition of the great arteries, tetralogy of Fallot, total anomalous pulmonary venous return and patent ductus
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1349	Describe the embryologic origin of coarctation of the aorta.
			SOM.MK.II.BPM1.3.CPR.2.ANAT.1350	Explain how coarctation of the aorta results in a difference between femoral and radial pulses and notching of the ribs.
Lecture 7	PHYS	Mechanics of the Cardiac Pump	SOM.MK.II.BPM1.3.CPR.2.PHYS.0096	Draw and describe the length-tension relationship in a single cardiac cell. Correlate length, tension, and velocity of shortening with the intact ventricle characteristics of end diastolic volume and pressure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0097	Define preload and explain why ventricular end-diastolic pressure, atrial pressure and venous pressure all provide estimates of ventricular preload. Explain why ventricular end-diastolic pressure provides the most reliable estimate.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0098	Define afterload and explain how arterial pressure increases or decreases afterload.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0099	Define contractility.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0358	Explain how the calcium transient differs between cardiac and skeletal muscle contraction.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0100	Describe the difference between cardiac performance and cardiac contractility and the impact of changes in preload, afterload, and contractility in determining cardiac performance.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0101	Explain how changes in sympathetic activity alter ventricular work, cardiac metabolism, oxygen consumption and cardiac output.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0102	Use the law of Laplace to predict the effects of pressure, radius and wall thickness on ventricular function.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0103	Illustrate and explain the phases and events of the cardiac cycle using a diagram of a ventricular pressure loop.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0104	Differentiate between the calculations and definitions of stroke volume and stroke work. Compare the stroke volume and stroke work using a pressure-volume loop.

Lecture 8	PHYS	Determinants of Myocardial Performance	SOM.MK.II.BPM1.3.CPR.2.PHYS.0105	Calculate ejection fraction from the values of end diastolic volume and end systolic volume and stroke volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0359	Describe the impact of changes in preload, afterload, and contractility in determining cardiac performance.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0344	Predict the change in ejection fraction that would result from a change in the a) preload, b) afterload, and c) myocardial contractility.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0106	Describe the timing and causes of the four heart sounds.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0107	Describe the expected auscultation sounds and pressure changes (cardiac cycle, pressure volume loops) that define mitral stenosis, mitral insufficiency, aortic stenosis, and aortic insufficiency.
Lecture 9	PHYS	Hemodynamics of Pressure, Flow and Resistance	SOM.MK.II.BPM1.3.CPR.2.PHYS.0108	Contrast flow and velocity and list the measurement units for each.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0109	Explain how Darcy's law maps the relationship between pressure, flow, and resistance in the vasculature and be able to calculate for one variable if the other two are known. Apply this relationship to the arteries, arterioles, capillaries, venules, and veins. Explain how blood flow to any organ is altered by changes in resistance to that organ.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0110	Discuss how Poiseuille's Law explains the changes in resistance with flow. Use it to calculate changes in resistance in a rigid tube (blood vessel). Explain the deviations from Poiseuille's law predictions that occur in distensible blood vessels.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0111	Discuss the relationship between flow, velocity and cross-sectional area. Determine the changes in these variables that occur with changes in vascular compliance.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0345	Contrast resistance and conductance in the vascular system and the effects of adding resistance in series vs. in parallel on total resistance and flow.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0112	Apply this information to solving problems characterized by a) resistances in series and b) resistances in parallel. Apply this concept to the redistribution of flow from the aorta to the tissues during exercise.
Lecture 10	PHYS	Hemodynamics of Systemic Circulation	SOM.MK.II.BPM1.3.CPR.2.PHYS.0113	List the factors that shift laminar flow to turbulent flow. Compare the relationship between velocity, viscosity, and audible events, such as murmurs and bruits.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0114	Describe single line blood flow and relate this to sickle cell anemia
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0115	Describe the relationship between wall tension, pressure and radius: La Place's law
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0116	Given systolic and diastolic blood pressures, calculate the pulse pressure and the mean arterial pressure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0117	Describe how arterial systolic, diastolic, mean, and pulse pressure are affected by changes in a) stroke volume, b) heart rate, c) arterial compliance, and d) total peripheral resistance.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0118	Describe how gravity affects blood flow and relate it to blood pressure gradient
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0119	Describe factors that affect venous return
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0120	Using Fick's Principle, discuss the dye dilution and thermodilution methods to calculate the cardiac output.
DLA 4	PHYS	Local Control of Blood Flow	SOM.MK.II.BPM1.3.CPR.2.PHYS.0121	Define autoregulation of blood flow. Distinguish between short-term and long-term autoregulatory responses and the mechanisms responsible for each.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0122	Describe how the theory of metabolic regulation of blood flow accounts for active hyperemia and reactive hyperemia.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0123	Describe the contribution of myogenic tone to blood flow regulation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0124	Identify the role of PO ₂ , PCO ₂ , pH, adenosine, and K ⁺ in the metabolic control of blood flow to specific tissues.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0125	Illustrate and explain the synthetic pathway for nitric oxide including substrate and the interplay between endothelium and vascular smooth muscle
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0126	Discuss the circumstances and the mechanisms whereby humoral substances contribute to regulation of the microcirculation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0127	Discuss the interaction of a) intrinsic (local), b) neural, and c) humoral control mechanisms and contrast their relative dominance in the CNS, coronary, cutaneous, and skeletal muscle vascular beds.
DLA 5	PHYS	Microcirculation	SOM.MK.II.BPM1.3.CPR.2.PHYS.0128	Differentiate the following terms: osmotic pressure, oncotic pressure, and hydrostatic pressure, as they pertain to movement across the endothelium of the capillaries.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0129	Based on the Starling hypothesis, explain how permeability, hydrostatic pressure and oncotic pressure influence transcapillary exchange of fluid
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0130	Describe how changes in capillary surface area affect the capacity for fluid exchange.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0131	Explain the Starling equation and how each component influences fluid movement across the capillary wall
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0132	Predict how altering pressure or resistance in pre- and post-capillary regions alters capillary pressure and the consequence of this change on transmural fluid movement.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0133	Discuss the lymphatics and how the structural characteristics allow the reabsorption of large compounds including proteins

			SOM.MK.II.BPM1.3.CPR.2.PHYS.0134	Contrast the structure of lymphatic capillaries and systemic capillaries, including the significance of the smooth muscle in the walls of the lymphatic vessels.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0135	Identify critical functions of the lymphatic system in fat absorption, interstitial fluid reabsorption, and clearing large proteins from the interstitial spaces.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0136	Discuss the relationship between the interstitial pressure and the lymph flow, and why an increase in interstitial pressure does not normally cause edema.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0137	Explain how edema develops in response to: a) venous obstruction, b) lymphatic obstruction, c) increased capillary permeability, d) heart failure, e) tissue injury or allergic reaction, and f) malnutrition.
Lecture 11	PHYS	Cardiac Output and Venous Return	SOM.MK.II.BPM1.3.CPR.2.PHYS.0138	Explain the cardiac function curve and the vascular function curve, and how factors such as afterload, myocardial contractility, blood volume, venous compliance, and total peripheral resistance modify these curves.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0139	Explain the concept of mean systemic filling pressure, its normal value, and the factors that can alter its value
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0140	Define venous return.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0141	Explain the vascular function curve and how changes in total peripheral resistance, blood volume, and venous compliance influence this curve.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0142	Explain why the intersection point of the cardiac function and vascular function curves represents the steady-state cardiac output and central venous pressure under the conditions represented in the graph.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0143	Explain how the intersection point of the cardiac and vascular function curves changes in certain situations such as exercise, use of vasopressor drugs, fluid infusion, acute hemorrhage, heart failure, and use of positive inotropic agents like digoxin
Lecture 12	PHYS	Cardiac Action Potential	SOM.MK.II.BPM1.3.CPR.2.PHYS.0164	Compare and contrast the ventricular myocyte action potential and the pacemaker action potential in terms of their phases and ionic basis.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0165	Explain the role of ion channels that contribute to various phases of the action potential of the ventricular myocytes and the pacemaker cells.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0166	Explain the causes of the long duration of the cardiac action potential, the resultant long refractory period, and the advantages of these characteristics for the cardiac muscle
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0167	Beginning in the SA node, diagram the normal sequence of cardiac activation (depolarization).
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0168	Explain why the AV node is the only electrical pathway between the atria and the ventricles, the functional importance of the slow conduction through the AV node, and the factors that influence conduction velocity through the AV node
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0169	Explain the characteristics of the pacemaker cells of the heart, the ionic mechanism of automaticity and rhythmicity, their location and spontaneous rate, and the neural and humoral factors that influence their rate
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0170	Compare the sympathetic and parasympathetic nervous system influence on heart rate and cardiac excitation and which division is dominant at rest and during exercise
DLA 6	PHYS	Special Circulations	SOM.MK.II.BPM1.3.CPR.2.PHYS.0144	Describe how the theory of metabolic regulation of blood flow accounts for active hyperemia and reactive hyperemia.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0145	Describe the phasic flow of blood to the ventricular myocardium through an entire cardiac cycle.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0146	Explain how arterio-venous O ₂ difference and oxygen extraction in the heart is unique when compared with other body organs.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0147	Explain the mechanism whereby the coronary blood flow is coupled to the myocardial workload and oxygen demand, and what type of stimuli can cause an increase in that flow
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0148	Explain the effects of the sympathetic nervous system stimulation on the heart rate, contractility, coronary vascular resistance, and coronary blood flow; and particularly during exercise, how the sympathetic stimulation has a direct and indirect effect determining the coronary blood flow
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0149	Explain the concept coronary vascular reserve, the role of collateral blood vessels, and the physiologic and pathological events that decrease the coronary vascular reserve
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0150	Discuss the mechanisms of regulation of the cerebral blood flow (myogenic, neurogenic, metabolic, and endothelial) and the importance of O ₂ , CO ₂ and pH in this process
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0151	Discuss the importance of intracranial pressure in the cerebral circulation
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0152	Contrast local and neural control of cutaneous blood flow.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0153	Discuss the unique characteristics of skin blood flow that are adaptive for body temperature regulation.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0154	Contrast the systemic and pulmonary circulations with respect to pressures, resistance to blood flow, and response to hypoxia.
Lecture 13	PHYS	Origin and Components of ECG	SOM.MK.II.BPM1.3.CPR.2.PHYS.0171	Explain the concept of dipole, vector of depolarization/repolarization, and how these vectors generated in the heart produce the waveforms of the ECG.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0172	Name the parts of a typical bipolar (Lead II) ECG tracing and explain the ECG waves, intervals, and segments in relation to the electrical state of the heart.

		ECG	SOM.MK.II.BPM1.3.CPR.2.PHYS.0173	Relate the ECG trace to the different parts of the cardiac action potential.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0174	Explain the electrocardiographic changes associated with the myocardial ischemia and cell death, and the concept of injury current and its relationship with the ST segment elevation or depression.
Lecture 14	PHYS	ECG Leads, Mean Electrical Axis and Interpretation	SOM.MK.II.BPM1.3.CPR.2.PHYS.0175	Explain the 12 leads used in standard ECG (I, II, III, aVR, aVL, aVF, V1-V6) including electrode placement, classification according to the plane and polarity, and lead axis.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0176	Explain why the ECG tracing looks different in each of the 12 leads.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0177	Discuss the concept of mean electrical axis of the heart, its normal range and the different methods to determine it.
DLA 7	PHYS	Applied ECG	SOM.1a.BPM1.3.CPR.3.PHYS.0178	Describe the alteration in conduction and ECG changes in most common arrhythmias: Sinus tachycardia, sinus bradycardia, A-V blocks, atrial fibrillation and ventricular fibrillation.
Lecture 15	BCHM	Chylomicrons, VLDL and LDL Metabolism	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0215	Discuss lipid transport in blood and the role of lipoproteins and albumin.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0216	Discuss chylomicron metabolism starting from formation to its uptake into the liver.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0217	Discuss the transport of dietary (exogenous) TAGs from the intestine to the peripheral tissues by chylomicrons
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0218	Describe the location and function of lipoprotein lipase in heart, skeletal muscle and adipocytes.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0219	Describe the conversion of VLDL into IDL and LDL and highlight the significance of lipoprotein lipase and hepatic lipase.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0220	Classify lipoproteins based on size and percentage of TAGs and cholesteryl esters.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0221	Outline and discuss uptake of LDL into liver and extra-hepatic tissues by receptor mediated endocytosis via the LDL receptor and remnant receptors.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0222	Describe the regulation of LDL-receptor synthesis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0224	Explain how oxidized LDL particles are formed and how they are involved with foam cell development and atherosclerosis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0230	Estimate LDL-cholesterol and VLDL-cholesterol using the Friedewald equation.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0233	Describe mechanism of action of statins and PCSK-9 inhibitors in hypercholesterolemia.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0234	Discuss the significance of Lp(a) and LDL-B as risk factors for coronary heart disease.
Lecture 16	BCHM	HDL and Lipoprotein Disorders	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0223	Discuss the functions and effects of deficiency of apo B-48, apo B-100, apo C-II, apo E.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0225	Describe reverse cholesterol transport and the role of HDL and SRB-1 receptors.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0226	Describe the synthesis, location and action of lecithin: cholesterol acyltransferase (LCAT) and activation by apo A-1.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0227	Describe the function of cholesterol ester transfer protein (CETP) and explain its role in cholesterol ester transfer from HDL to VLDL, IDL and LDL.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0229	Describe hypoalphalipoproteinemia (Tangier disease) and abetalipoproteinemia and hypobetalipoproteinemia (molecular mechanism, lipid profile abnormalities, basis for clinical features).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0231	Differentiate different hyperlipidemias (Type I, IIa, IIb, III, IV and V) using lipid profile (Serum TAG and Serum cholesterol), lipoprotein electrophoresis, clinical features and molecular mechanisms and inheritance patterns.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0232	Review the other names for the hyperlipidemias type I-V.
DLA 8	BCHM	Nutritional Aspects of Cardiovascular Disease	SOM.MK.III.BPM1.1.CPR.2.BCHM. 0235	Describe the major risk factors of cardiovascular disease.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0236	Describe risk factors for CHD related to abnormal lipoprotein pattern.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0237	Discuss amount and types of dietary fats related to CHD (Omega-6 and omega-3 fatty acids)
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0238	List the diagnostic criteria for obesity: BMI, waist circumference, waist/hip ratio
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0239	Discuss the metabolic effects of obesity and the criteria for identification of metabolic syndrome (insulin resistance syndrome).
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0240	Discuss the metabolic effects of weight loss.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0241	Review the role of dietary risk factors for development of coronary heart disease (CHD) like high salt and high glycemic index.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0242	Highlight role of protective factors against development of coronary heart disease (CHD) like fiber, moderate alcohol, and vitamins that decrease homocysteine levels.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0243	Review the role of antioxidants, vitamins and their long-term benefits.
			SOM.MK.III.BPM1.1.CPR.2.BCHM. 0244	Describe benefits of DASH diet and Mediterranean diet.
DLA 9	BCHM	Eicosanoids	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0252	Describe the synthesis of the eicosanoid molecules, their general structure, and their precursors. Indicate inhibitors of eicosanoid synthesis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0253	Discuss the role of eicosanoids in the regulation of inflammation (PGE ₂), gastric acidity, blood flow, and platelet aggregation (PGI ₂ and TXA ₂).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0254	Compare and contrast the effects of TXA ₂ vs TXA ₃ and their effects on vascular health.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0255	Discuss the role of leukotrienes, cysteinyl leukotrienes and SRS-A in asthma and in anaphylactic shock

Lecture 17	BCHM	Primary and Secondary Hemostasis	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0245	Outline the four phases of hemostasis.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0246	Discuss vascular spasm and the role of endothelin.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0247	Describe platelet plug formation.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0248	Explain the role of integrins GPIb, GPIa, GPIIb/IIIa and the role of von Willebrand factor in platelet plug formation
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0249	Describe the role of platelet-activating factors (ADP, TXA ₂) and the changes in platelets on activation.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0250	Describe the coagulation cascade, including the extrinsic, intrinsic and common pathways.
Lecture 18	BCHM	Fibrinolysis and Disorders of Hemostasis	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0251	Indicate the role of calcium in hemostasis
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0256	Discuss the role of vitamin K in the post-translational modification of clotting factors
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0257	Predict the clinical manifestations in a patient with vitamin K deficiency
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0258	Outline the anticoagulant and fibrinolytic roles of PGI ₂ , antithrombin III, proteins C and S, and plasmin.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0259	Appraise the role of bleeding time, platelet function tests (ristocetin cofactor assay), prothrombin time (INR), and activated partial thromboplastin time (APTT) in the differentiation of bleeding disorders.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0260	Indicate the mechanism of action of aspirin, warfarin, Tissue plasminogen activator and heparin
Lecture 19	HCB	Histology of Blood Cell Formation	SOM.MK.I.BPM1.3.CPR.1.HCB.1157	List the functions of blood.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1158	Describe the method of preparing peripheral blood smear.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1159	Describe the cellular components of blood.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1160	Describe the extracellular components of blood.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1161	List the formed elements of blood.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1162	Give the function of erythrocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1163	Describe the microscopic structure of erythrocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1164	Describe changes in erythrocytes in anemia and polycythemia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1165	Describe changes in erythrocytes in polycythemia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1166	State the hereditary molecular defects in the erythrocyte cell membrane that leads to hereditary spherocytosis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1167	Describe the genetic changes involved in the hemoglobin molecule resulting in thalassemia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1168	Describe poikilocytosis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1169	Describe anisocytosis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1170	Define hematopoiesis (hemopoiesis).
			SOM.MK.I.BPM1.3.CPR.1.HCB.1171	List the 3 major organs involved in hematopoiesis before birth.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1172	List the sites where hematopoiesis is predominant as age advances.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1173	Describe the characteristics of a stem cell.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1174	Define totipotent, pluripotent and multipotent stem cells.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1175	Describe hematopoietic stem cell (HSC).
			SOM.MK.I.BPM1.3.CPR.1.HCB.1176	Describe hematopoietic stem cell niches.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1177	Describe the cellular organization, vascularity and functions of bone marrow.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1178	Describe the role of growth factors, cytokines and hormones involved in hematopoiesis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1179	Describe the process of erythropoiesis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1180	Describe the stages of erythrocyte differentiation.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1181	Describe changes in erythrocytes in sickle cell anemia.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1182	Briefly describe the role of iron, vitamin B ₁₂ and folic acid in the development of erythrocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1183	Describe the effects of renal failure on erythropoiesis.
Lecture 20	HCB	Histology of Blood Cell Function	SOM.MK.I.BPM1.3.CPR.1.HCB.1184	Describe the myeloid and lymphoid lineages.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1185	List all the leukocytes and their subdivisions.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1186	List the components of azurophilic granules and their functions.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1187	Give the function of neutrophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1188	Describe the microscopic structure of neutrophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1189	Discuss the function of the products secreted from the specific granules of neutrophils
			SOM.MK.I.BPM1.3.CPR.1.HCB.1190	Give the function of eosinophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1191	Describe the microscopic structure of eosinophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1192	Discuss the function of the products secreted from the specific granules of eosinophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1193	Give the function of basophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1194	Describe the microscopic structure of basophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1195	Discuss the function of the products secreted from the specific granules of basophils.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1196	Describe the process of granulopoiesis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1197	Describe the process of lymphopoiesis
			SOM.MK.I.BPM1.3.CPR.1.HCB.1198	State the function of lymphocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1199	Describe the microscopic structure of lymphocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1200	Describe the process of monopoiesis
			SOM.MK.I.BPM1.3.CPR.1.HCB.1201	Give the function of monocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1202	Describe the microscopic structure of monocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1203	Describe the white blood cell count and give an example of clinical conditions that may increase or decrease the cell count for each WBC.

			SOM.MK.I.BPM1.3.CPR.1.HCB.1204	State the life span of each formed element of blood.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1205	Describe the histological structure of thrombocytes.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1206	Describe the role of platelets in hemostasis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1207	Describe the process of thrombopoiesis.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1208	Describe the structure and function of megakaryocytes and formation of platelets.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1209	Histologically identify the following precursors: reticulocytes, band cells and megakaryocytes in the normal development of erythrocytes, granulocytes and platelets respectively.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1210	Describe the process of bone marrow aspiration & biopsy techniques and their clinical significance.
			SOM.MK.I.BPM1.3.CPR.1.HCB.1211	Compare normal bone marrow vs. aplastic anemia and leukemia in a peripheral smear.
Lecture 21	HCB	Histology of Blood Cell Activation and Immune Responses	SOM.MK.1.BPM1.3.CPR.1.HCB.1212	List the functions of the lymphatic system
			SOM.MK.1.BPM1.3.CPR.1.HCB.1213	Describe the morphology (including the activation process) of the following blood cells: CD4+ & CD8+ T cells, B cells & plasma cells and natural killer cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1214	Identify the function(s) of the following blood cells: CD4+ & CD8+ T cells, B cells & plasma cells and natural killer cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1215	Describe the morphology (including the activation process) of antigen presenting cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1216	Identify the functions of antigen presenting cells.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1217	List the two types of immune response.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1218	Describe the two types of immune response.
			SOM.MK.1.BPM1.3.CPR.1.HCB.1219	Using your knowledge of blood cells, demonstrate the link between the immune system and inflammation
Lecture 22	BCHM	Pentose Phosphate Pathway: Significance in RBC and WBC	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0262	List the tissues in which the PPP is active, and identify its subcellular location
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0263	Summarize the function of the PPP (metabolic roles of pentoses and NADPH).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0264	Explain how malaria may be a factor in the relatively high rate of G6PD deficiency alleles in some populations
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0265	State the reactions of the oxidative portion of the PPP (emphasize the significance of G6PD)
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0266	Provide an overview of the intermediates of the non-oxidative phase of the PPP highlighting the importance of transketolase.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0267	Explain how the PPP is regulated and how different parts of the pathway are used, depending on the needs of the cell.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0268	Evaluate the use of NADPH in: Detoxification of hydrogen peroxide, Phagocytosis, Nitric oxide synthesis, Reductive biosynthesis
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0269	Describe glucose 6-phosphate dehydrogenase (G6PD) deficiency including precipitating factors, X-linked recessive inheritance, symptoms, laboratory findings, biochemical basis of findings (hemolysis and Heinz bodies in RBC).
DLA 10	BCHM	Reactive Oxygen Species	SOM.MK.I.BPM1.1.CPR.2.BCHM. 0270	Define the terms radical, free radical and reactive oxygen species (ROS). Discuss cellular damage and effects of ROS.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0271	Describe formation of superoxide anion radical related to the ETC or cytochromes.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0272	Describe the reactions catalyzed by superoxide dismutase, catalase and glutathione peroxidase (selenium containing) and indicate the ROS scavenged.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0273	Discuss the formation of hydroxyl radicals related to ionizing radiation, Haber- Weiss reaction and Fenton reaction.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0274	Discuss the compartmentation of free radical defense in the cell.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0275	Indicate non-enzymatic radical scavengers. Include vitamin C and E (found in fruits and vegetables).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0276	Indicate the biochemical defect in amyotrophic lateral sclerosis (ALS).
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0277	Discuss the beneficial role of ROS formation as an immunological response in phagocytosis and describe the reactions catalyzed by NADPH oxidase and myeloperoxidase.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0278	Discuss the direct and potential effects of excess nitric oxide formation. Discuss the importance of inducible nitric oxide synthase for formation of RNOS.
			SOM.MK.I.BPM1.1.CPR.2.BCHM. 0279	Predict the effect of NADPH oxidase deficiency and myeloperoxidase deficiency. Indicate the biochemical defect in chronic granulomatous disease.
DLA 11	PHYS; HPWP	Exercise Physiology	SOM.MK.II.BPM1.3.CPR.2.PHYS.0186	Describe the cardiac output, circulation, respiration response to physical exercise.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0187	Explain the adaptations to training in the cardiovascular, respiratory, musculoskeletal systems and metabolic adaptations
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0155	Describe the cardiovascular consequences of exercise on peripheral resistance, and cardiac output
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0156	Contrast the neural and local control of skeletal muscle blood flow at rest and during exercise.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0157	Discuss the interaction of neural and humoral control mechanisms and contrast their relative dominance in the skeletal muscle vascular beds
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0188	Discuss the benefits of regular physical activity and exercise.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0189	Discuss the risks of exercise
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0190	Describe the medical evaluations necessary prior to exercise

			SOM.MK.II.BPM1.3.CPR.2.PHYS.0337	Explain the importance of health education and promotion in health care as well as within culturally diverse populations.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0338	Demonstrate appropriate techniques for patient education regarding exercise recommendations based on WHO guidelines
Lecture 23	PHYS	Hypertension & Heart Failure	SOM.MK.II.BPM1.3.CPR.2.PHYS.0179	Define and classify hypertension.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0180	Define acute, chronic and congestive cardiac failure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0181	Describe the compensatory responses to heart failure and explain the outcomes of advanced heart failure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0182	Describe the major cardiovascular changes that occur with ageing (i.e. compliance changes).
DLA 12	PHYS	Control of Blood Pressure	SOM.MK.II.BPM1.3.CPR.2.PHYS.0339	Discuss rapid control of blood pressure by baroreceptor reflex.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0340	Construct a step-by-step flow chart to indicate changes in the cardiovascular system that occur when the blood pressure changes in the body, taking examples of increase and a decrease in blood pressure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0341	Discuss long-term regulation of blood pressure by Renin-angiotensin-aldosterone mechanism.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0342	Discuss the role of ADH (vasopressin) in control of blood pressure.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0343	Discuss other mechanisms of control of blood pressure such as low pressure baroreceptors and Atrial natriuretic peptide.
Lecture 24	PHYS	Hemorrhagic Shock	SOM.MK.II.BPM1.3.CPR.2.PHYS.0183	Describe the direct cardiovascular consequences on cardiac output, central venous pressure, and arterial pressure after a loss of 30% of the circulating blood volume.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0184	Describe the compensatory mechanisms activated by these changes. Explain the positive feedback mechanisms activated during severe haemorrhage that may lead to circulatory collapse and death.
			SOM.MK.II.BPM1.3.CPR.2.PHYS.0185	Contrast the change in plasma electrolytes, hematocrit, proteins, and oncotic pressure following resuscitation from haemorrhage using a) water, b) 0.9% NaCl, c) plasma, and d) whole blood.